RealQM vs Conduction:
the Free Boundary as Carrier

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Abstract

RealQM models matter as non-overlapping unit charge densities in ordinary three-dimensional space, one per electron, meeting at free boundaries and minimising the Coulomb energy. This article asks whether such a construction **conducts**, and reports that the question cannot be answered in this formulation — for reasons that turn out to be structural rather than numerical, and that locate the boundary of the framework's scope.

What would move is not an electron. An electron here is a unit of charge density on a domain; what exists is the *partition* of space, and the carrier is a **free boundary** — in the relation a dislocation bears to the slip it produces, charge arriving at the far end while nothing material crosses the sample. That identification is the article's positive content: it accounts for every null result reported below, each of which had imposed a carrier obliged to translate bodily through occupied space, and it is why the framework's one successful transport calculation is Grotthuss conduction, where the carrier is a proton. It is not, however, the mechanism of ordinary conduction: perfect crystals conduct *best*, and a defect-mediated carrier would predict the opposite. The free boundary is what a formalism with real densities is restricted to, not what carries current in a metal.

The activation energy for such a boundary to advance one site is not extractable here, and the demonstration is the methodological contribution. Scanning the carrier through a *full lattice period* rather than sampling two configurations supplies a control that costs nothing: translation by one spacing must return the same energy, so the mismatch is a per-run noise floor measured alongside the quantity it bounds. Of eight such scans, varying core radius and lattice spacing, **six fail that control**, and the two that pass survive no change of either parameter. The one scan in which both parameters are determined rather than chosen — r_c from the atom's ionisation energy, the spacing from that radius's own equilibrium — fails by the widest margin. An earlier draft quoted $E_a = 555$ meV and drew a verdict from its position relative to 500; that figure is withdrawn, along with the length-independence and the comparisons built on it. Two artifacts are identified in its place: a drift of the carrier toward a chain end, and grid-locking of a staircase interface, each of order an electron-volt and each exactly reproducible, which is why they passed so long for signal.

Underneath the arithmetic lies the structural obstacle. Non-overlap makes the transfer integral vanish identically, so band conduction is absent in kind and not by degree; and in this scheme nuclei move while electrons *relax to a static minimum*, so there is no electron equation of motion and hence no object corresponding to a current. A minimisation has no time in it. Attempting to infer transport from an energy landscape is a proxy, and it is the proxy that

breaks. Electronic conduction therefore lies outside static RealQM, while protonic and ionic conduction lie inside it; a conduction calculation belongs in the time-dependent form where a real density carries a complex current.

Two further results bound the framework. The interface coefficient vanishes exactly, $C = 0$, removing degeneracy pressure and leaving a vanishing screening length and a dispersionless plasmon; a Robin condition at $\beta a = 1.146$ restores $C = C_{\mathrm{TF}}$, but β that large *unbinds* H_2, which fails by $\beta a \approx 0.71$. And on the same chains, exact diagonalisation in a minimal basis puts the carrier on four or five sites where this construction pins it to one — the framework errs by **over-localising**, which is the direction $U \to \infty$ implies, so it fails as its own analysis says it should rather than arbitrarily.

1 The extension, and the object it is made for

In this article we extend RealQM, developed for atoms and molecules, to **two-component systems**: ions and electrons, both charged, both free to move, interacting by Coulomb forces and nothing else. A metal is such a system, and the question pursued here is the elementary one such a system poses — whether it *conducts*, and if so what carries the charge. The same construction covers a stellar interior and the recombination era of a Coulomb cosmology, which are treated as the classical plasma, where $\omega_p^2 = 4\pi n$ comes out exactly.

RealQM [1, 2] represents an N-electron system by N non-negative densities ψ_i, each carrying unit charge on its own domain Ω_i, the domains non-overlapping and their interfaces free to move to where the energy is least. The energy minimised is Coulomb's:

$$E[\{\psi_i\}, \{\Omega_i\}] = \sum_i \int_{\Omega_i} \Big(\tfrac{1}{2} |\nabla \psi_i|^2 - \sum_a \frac{Z_a}{|x - x_a|} \psi_i^2 \Big) dx + \tfrac{1}{2} \sum_{i \neq j} \iint \frac{\psi_i^2(x) \psi_j^2(y)}{|x - y|} \, dx \, dy, \qquad (1)$$

subject to $\int_{\Omega_i} \psi_i^2 = 1$. There is no wavefunction in $3N$ dimensions, no antisymmetry and no self-interaction; the role played by the exclusion principle is played by the requirement that domains do not overlap. And — of particular relevance here — there is no smeared continuum anywhere in the construction: everything in it is a discrete unit charge occupying a region of ordinary space.

A two-component system of ions and electrons is therefore not an idealisation the theory must be adapted to describe. It is what the theory is already made of, in a new arrangement.

Every previous test of this construction — ionisation energies across the periodic table, molecular binding, nuclear binding by dual Coulomb confinement [3] — has been on *bound* matter, with an attractor always present. A plasma is the first case in which large numbers of charges of both signs are free to arrange themselves, and it is a fair question what the framework then gives.

2 What kind of conductor this can be

The construction does not have a free choice of target, and identifying it correctly settles most of what follows. One electron per domain, non-overlapping, is usually described as the Hubbard model at $U \to \infty$: double occupancy forbidden, every site singly occupied. That is right as far as it goes and it is not far enough. The Hubbard Hamiltonian has two parameters, and non-overlap fixes both. $U = \infty$ because two electrons may not share a domain; but also $t = 0$, because the transfer integral *is* the overlap of neighbouring orbitals and there is none. The construction is therefore not the Mott limit but the **atomic limit**, and the difference is not a refinement:

	$U \to \infty$, $t \neq 0$	$U \to \infty$, $t = 0$
half filling	Mott insulator	isolated atoms
superexchange	$J = 4t^2/U$	none
doped	**conducts**	still nothing
spin dynamics	yes	none

A Mott insulator is insulating but alive: it carries magnetic exchange, and doping it produces a conductor, which is what the cuprates are. The atomic limit is a collection of atoms with electrostatics between them. This accounts for every negative result below, and for one in particular: doping a Mott insulator works *through* t, so at $t = 0$ there is nothing for doping to activate — which is exactly what the measurement shows, the barrier rising with carrier concentration rather than falling. Metals are therefore out of scope in kind and not by degree. Their carriers are extended Bloch states, their transport is band motion, and their defining measurements — de Haas–van Alphen oscillations, a linear electronic specific heat, positive plasmon dispersion — belong to a degenerate electron gas with a sharp Fermi surface, which no arrangement of densities on domains appears able to supply. Two results usually read as failures follow from the scope rather than against it: $d\sigma/dT > 0$ is *correct* for an activated conductor and fatal only for a metal, and the absence of a Fermi surface does not bear on a dilute carrier gas, which is non-degenerate and has none to reproduce.

What remains within reach is matter whose carrier is a localised charge on a site, whose gap is on-site repulsion rather than band structure, and whose transport is activated hopping. That is a real class — doped Mott insulators, organic and molecular semiconductors, polaronic oxides, amorphous semiconductors — and this article set out to claim it. §4 measures the activation energy and finds 613–1723 meV, against the 10–500 meV those materials show. The claim does not survive its own measurement, and what the construction describes is the insulating end of the class and not the conducting one. Silicon and the III–Vs were never in scope, their gaps being band gaps.

3 The interface, and the one number that is wrong

The free interface has a defect of its own, recorded here because it conditions everything the partition does, though the hopping barrier does not turn on it. Partitioning a gas of density n gives a kinetic energy density $t(n) = C\,n^{5/3}$ — the Thomas–Fermi *form*, obtained from partition geometry rather than Fermi statistics — but the flat trial function is admissible on a free boundary, so $C = 0$ exactly. There is no degeneracy pressure, the screening length vanishes rather than being finite, and the plasmon does not disperse. The defect is one number and not the construction: C is fixed by the interface condition, and a Robin condition at $\beta a = 1.146$ gives $C = 2.871 = C_{\mathrm{TF}}$ exactly, against Dirichlet's 14.804, all three carrying the correct $n^{5/3}$ scaling.

Whether that value is tolerable for matter is the open question, and it is sharper than a parameter check. Adding the surface term and scanning it, H_2 unbinds by $\beta a \approx 0.71$, below the 1.146

the electron gas requires: Teller's theorem, that Thomas–Fermi binds no molecule, is met inside the framework rather than inherited from outside it. Double occupancy — the repair conduction would need at half filling — becomes favourable at $\beta a = 0.451$ and is affordable by comparison. These figures were obtained at differing densities and are not strictly commensurable; the equation of state is pursued elsewhere, and what matters here is only that the interface carries no cost of its own unless one is supplied.

A caution about specified partitions, learned by getting it wrong. The method of §4 evaluates the energy of a *stated* partition, and it is sound only when the stated partition is one the physics admits. The interface is not a surface to be shaped at will: the wavefunction is a single global field partitioned by labels, continuous across every interface, each domain carrying one unit of charge, and the boundary sits where the densities balance. Freezing a partition removes that last condition. An attempt to measure a bending rigidity by imposing a curvature, freezing it and relaxing the fields gave an energy falling symmetrically and quadratically in the imposed curvature, which reads convincingly as a negative rigidity; but released, a boundary started from the curved configuration returns to the flat one. The flat interface is the balance surface and is stable, and what the frozen scan measured was the energy gained by *dropping the balance condition.* A specified partition gives a legitimate energy difference between configurations the physics admits — as symmetry guarantees for the two in §4 — and a meaningless one along a direction it forbids. Nothing in the numbers distinguishes the two cases; the check is to release the boundary and see whether it stays.

4 Computing with it anyway

Electric conduction is the interaction of the electrons with the ion lattice. It is the elementary case — ordinary matter at ordinary temperature — and it is for that reason the *demanding* one here, since ordinary-temperature conduction is degenerate matter, and degeneracy is what this framework has least of.

One property of the construction is needed before anything else, and it is the property that makes conduction a live question rather than a hopeless one. With the free interface used throughout, a uniform density is admissible on any partition: $\psi_i \equiv |\Omega_i|^{-1/2}$ satisfies the normalisation, minimises the electrostatic energy against a neutralising background, and has $\nabla\psi_i = 0$. Redrawing the boundary therefore costs nothing,

$$\text{a free interface between domains of equal density carries no confinement energy,} \tag{2}$$

and the boundary is the only object this framework moves — its solver evolves each interface by the local density balance $(\rho_i - \rho_j)/(\rho_i + \rho_j)$, transferring ownership of a cell when the neighbour's density exceeds its own. Domains do not translate; they are redrawn. *A vanishing interface cost is a vanishing cost of transport.* What that same fact does to the equation of state is a separate matter and is taken up in §3, where it is much less welcome.

Metallic conduction is the paradigm of *delocalisation*: the carriers are extended states spanning the crystal, whereas RealQM's electrons are localised by construction, each in its own domain, with non-overlap requiring N carriers to occupy N disjoint regions. The natural minimiser of (1) is a set of compact cells, which describes a Mott insulator more naturally than a conductor. Nor is there an evident mechanism for the metal–insulator distinction, which in standard theory is band filling — two electrons per band per cell — an exclusion-principle count the framework does not have; and with no Fermi surface, the carrier density in $\sigma = ne^2\tau/m$ has no referent.

What does fit naturally is the opposite regime: localised charges hopping between domains by thermally activated jumps, as in semiconductors, disordered systems and small-polaron transport. That needs no Fermi surface and follows from the same thermal motion that supplies the pressure.

This yields a prediction crude enough to be decisive, and it concerns only a sign. Metallic conductivity *falls* with temperature; hopping conductivity *rises*. If RealQM transports charge by hopping rather than band motion it should give $d\sigma/dT > 0$. Within the scope set in §2 that is the *right* answer, activated transport being what a localised-carrier semiconductor does; it would be fatal only for a metal, which is why the target matters. Whether it does has not been computed for electrons, and the pessimistic reading should be held loosely: the same lattice that reproduces alkali lattice constants is present here too.

For *protons* the mechanism has already been demonstrated in this framework. Grotthuss conduction — an excess proton migrating along a hydrogen-bonded chain of water molecules by successive transfers, rather than by bodily motion of any one particle — is hopping transport in its purest form, and it is a standard RealQM calculation: a chain of waters with O–O spacing near 2.65 Å, an excess proton, and the question whether it hops from one molecule to the next. Related transfers, $\mathrm{HF} + \mathrm{H_2O} \to \mathrm{F^-} + \mathrm{H_3O^+}$ and $\mathrm{HCl} + \mathrm{NH_3} \to \mathrm{Cl^-} + \mathrm{NH_4^+}$, are computed the same way.

This matters for the argument rather than merely illustrating it. Charge transport by transfer between localised domains is not a regime the framework must be stretched to reach — it is one it already handles, in the case where the carrier is a proton.

The electron case cannot presently be tested in the same way, and the obstruction is instructive enough to be worth setting out. Two chain calculations were attempted here — five hydrogens with an excess electron, and five hydrogen kernels with one electron missing, the electronic counterparts of the proton wire — and both returned nothing. Neither result means what it appears to.

The cause is not what it first appeared. Kernel proximity assigns the labels only *initially*; thereafter a free-boundary rule evolves each interface from the local density balance and transfers ownership of a cell when the neighbouring domain's density exceeds its own. Partitions therefore do rearrange, and electron transfer is representable in principle. What defeated the attempts here was specific to each: a carrier given no kernel feels no force from the lattice; a bare kernel is given no domain, so a vacancy has nothing to migrate into; and a carrier placed on an occupied site can seize the nuclear region outright, since with free interfaces no confinement cost opposes it. Each is a property of the construction, not of (1).

The deeper cause survives that repair. Conduction is charge in motion, and (1) is an energy minimisation: it returns the equilibrium partition, not a current. Given complete freedom in the labels one would still obtain the ground state, and for a symmetric chain that state places one domain per site by symmetry, with nothing flowing. A minimisation has no time in it.

Which filling the measurements belong to, and which needs the repair. It is worth being exact about this, because it determines how much of the article depends on β. At *half filling* — one electron per site, every domain occupied — the $U \to \infty$ construction forbids conduction outright: there is no vacancy for a carrier to enter, and the insulator is structural rather than dynamical. That is the case the double-occupancy repair is for. But it is not the case measured here. The chains reported above are doped by ionising every second atom, leaving half the sites empty, and every barrier in this article is the cost of a carrier moving into a *vacancy*. Single occupancy handles that without amendment.

So the transport results stand on their own: they require no finite U, no double occupancy and no β. What requires the repair is the undoped, half-filled chain, which is not run here. This is the stronger position of the two — the measurements are independent of the proposal made to extend them — and it locates precisely what the proposal would buy: not better hopping, but conduction at a filling where the construction currently permits none.

What the framework does possess is an asymmetry of dynamics. Nuclei move, under classical equations of motion; electrons relax, to a static minimum. Charge carried by *nuclei* is therefore representable — which is precisely why the Grotthuss calculation works, the carrier there being a proton — and charge carried by electrons is not, because in this scheme electrons have no equation of motion at all. The correct machinery exists elsewhere in the programme, in the time-dependent form where a real density carries a complex current, and a conduction calculation belongs there rather than here.

What the barrier means as a field. A barrier becomes a threshold field when divided by the charge and the hop distance. Taking the range of §5 rather than any single value, E_a/ea runs from 0.07 to 0.9 V per lattice site at $a = 6.0$; copper carrying an ampere runs at 4×10^{-12} V per site. The comparison spans an order of magnitude and is uncertain by that much, but ten orders separate the two either way, so the conclusion — that no field a material survives will drive this by direct depinning — does not depend on where in the range the barrier falls.

Why a chain rather than a ring. A ring supplies a clean barrier — no ends, every site equivalent, the energy obliged to repeat with the lattice period — but it cannot host the drive: a conservative field has zero circulation, so no bias drives a sea around a closed loop. A chain has ends, a potential drop along it is an ordinary thing to apply, and a single carrier may enter at one end and leave at the other, which is the transport event itself.

The partition must be specified, not searched for. Before the number, the method, because the obvious approach fails and the failure is instructive. A kink is *not* the ground state, so a free-boundary minimisation relaxes it away — asked to hold one, the solver oscillates between the kinked and uniform partitions and the energy never settles, wandering over 0.05 Ha against a barrier of order 0.01. That is not a convergence failure to be tuned away: it is unaltered by the surface tension β, by reducing the boundary step fivefold, and by freezing the labels outright.

The cause is general and it disqualifies the free boundary for every transport quantity in this article. The interface moves by *local single-cell label flips*, so a run descends to whichever configuration no single flip improves — the nearest stuck point, not the minimum — and which one that is depends on the initial condition. Four independent geometries agree:

geometry	spread	against a barrier of
ring, between two builds	0.005 Ha	0.012–0.022
kink chain, across steps	0.05 Ha (never settles)	~ 0.01
$H^- + H$, four seeds	0.031 Ha	~ 0.001
alkali chain, two seeds	0.013 Ha	~ 0.001

In every case the ambiguity exceeds the quantity sought, by one to two orders. It is a failure of the *search*, not of the energy functional, which is why nothing in the functional touches it. It is also why the ring's 0.01186 Ha moved to 0.022 Ha under a recompilation that altered no physics: rounding at the last bit flips cells sitting near the threshold, and a different stuck point follows. The rigid-sliding figure should be read as a bound of that order rather than a determination, and the doping barriers of the earlier barrier scans, obtained the same way, with it.

The repair is not a better minimiser but no minimiser. The domain walls of a chain are planes perpendicular to the axis, so they can be *placed*: specify the partition, freeze it, and relax only the wavefunctions. Nothing is searched for, the same input returns the same output, and the question asked is no longer "where does the boundary go" but "what does this configuration cost" — which is the question a barrier actually poses. It is the device that makes the H_2 midplane exact, applied to a coordinate instead of a symmetry.

5 The barrier is not extractable, and the control that shows it

A barrier taken as the difference of two configurations is only as good as the cancellation between their errors, and that cancellation is not something one may assume. The check is available and costs nothing conceptually: scan the carrier's position through a *full lattice period* rather than sampling two configurations, and require the scan to return to where it began. On an infinite chain $E(c{=}0)$ and $E(c{=}1)$ are the same configuration translated by one spacing and must agree exactly. Their mismatch is therefore a per-run noise floor — finite size and discretisation together — measured in the same run as the quantity it has to be compared against.

Two further benefits come with it. The barrier becomes the *amplitude* of $E(c)$, so nothing depends on having guessed where the extrema lie; and the shape of $E(c)$ is visible, so a genuine corrugation can be told from a monotone drift. Both matter, as it turns out: the maximum is generally not at the half-way point that a two-configuration measurement assumes.

Applying it is decisive. Eight scans, each nine configurations over one period, with the core radius and the lattice spacing varied:

a (a_0)	r_c	amplitude	control	ratio	
6.0	1.60	770 meV	699 meV	1.1	fails
6.0	1.60	1087	535	2.0	fails
4.5	0.80	1854	1524	1.2	fails
4.5	1.20	1735	936	1.9	fails
4.5	1.60	1900	78	24.4	passes
4.5	1.93	2597	584	4.4	passes
5.5	1.93	1949	1192	1.6	fails

Six of the eight fail their own control, and the two that pass do not survive a change of either parameter. Worse for any attempt to rescue a number from them, the passes cluster at one spacing, $a = 4.5$, across three different core radii — including one for which 4.5 is not the equilibrium. Had the control been passing because a bound chain holds the carrier, the calibrated radius at *its own* equilibrium would have passed; it fails by the widest margin in the table.

The last row is the one that settles the matter. Everywhere else in this article at least one parameter was chosen for convenience: $r_c = 1.6$ because it was to hand, $a = 6.0$ because the first scans used it. In that row both are determined — $r_c = 1.93$ from the isolated atom's ionisation energy, and $a = 5.5$ from that radius's own equilibrium, located by a five-point scan of the defect-free chain with a minimum at 5.43 a_0. *At parameters the framework does not get to choose, no barrier can be quoted.*

Why, in one line of arithmetic. The totals converge only to first order in the mesh, at about 0.91 Ha per unit h, so at $h = 0.30$ each still carries some 0.27 Ha of discretisation error — a fifth of the total — while the barrier differenced from the pair is 0.01 Ha. The quantity sought is under four per cent of the error in each quantity it is subtracted from, and would need better than ninety-six per cent cancellation to survive. It does not get it: refining the mesh alone at fixed everything else moves the value $315 \to 371 \to 552 \to 265$ meV, without converging and without monotonicity. Brute force cannot close this. Reducing the error merely to the size of the barrier needs $h \approx 0.011$, a 4200^3 grid; reaching ten per cent of it needs 42000^3.

Two artifacts, both identified. What the scans do contain is a monotone drift of 0.5–1.5 eV per site, the carrier being attracted to a chain end, superposed on a deterministic wiggle of similar size. The wiggle is grid-locking, and it can be caught in the act: at $h = 0.348$ the configurations $c = 0.125$ and $c = 0.250$ return *identical* energies to five decimals, because the cell-by-cell assignment did not flip between them. The partition is a staircase pinned to the grid, and the interface energy jumps as the staircase does. Neither artifact is noise — repeated runs of one configuration agree to all five decimals recorded — which is precisely why scatter of this kind survived so long undetected: it is exactly reproducible, and reproducibility was taken for correctness.

A caution about spacing, from the exact side. Every barrier first computed here used $a = 6.0$. On a hydrogen chain of the same geometry, full configuration interaction puts the binding at 0.8 mHa — 21 meV for six atoms, against the same atoms separated. *The chain is not bound* *at that spacing.* A lattice that does not hold together cannot pin a carrier, so the absence of a periodic corrugation there is the expected result and not a finding about RealQM. This invalidates the original comparison with hopping semiconductors on its own, before any question of numerical resolution arises: those materials are bound solids.

What would be needed instead. Not a finer grid. The failure is structural to the method of differencing large totals, so the routes out change what is computed. Either the barrier is obtained as a path integral of the interface force, $dE/d\lambda = \oint_\Gamma [w]\, v_n\, dS$, whose error scales with the interface rather than with the volume; or the partition is given sub-cell resolution so that E becomes a smooth function of where the walls lie instead of a rough one. Both are projects rather than adjustments, and neither was attempted here.

The control is sound, and a ring shows it. Lest the failures above be read as a defect of the specified-partition method itself, the control was checked on a geometry whose symmetry is exact. A ring of six kernels carrying six electrons, the partition rotated rigidly by ϕ in units of the spacing, must return the same energy at $\phi = 1$ as at $\phi = 0$ — the two are related by a relabelling — and must be symmetric about $\phi = \frac{1}{2}$. It is, to every digit recorded: $E(0) = E(1) = 0.56101$, $E(0.125) = E(0.875) = 0.35258$, $E(0.375) = E(0.625) = -0.21304$, with one pair disagreeing by 0.011 Ha and thereby setting the noise floor. So the method reproduces exact translations exactly, and the chain's asymmetry — its minimum displaced from $d = \frac{1}{2}$, which a reflection through a core forbids — is the finite chain's two end domains, 29% of a seven-core system, and not the partition. That ring is not otherwise usable: its angular sectors widen with radius, most of each domain lies far from its kernel, and the total energy comes out *positive* for a system six hydrogen atoms would bind at -3 Ha. It settles the control and nothing else.

Why this quantity in particular resists measurement. The construction is a Frenkel–Kontorova system, and saying so explains the difficulty rather than merely naming it. The electron domains are a chain with elastic stiffness κ supplied by Coulomb repulsion between neighbours; the kernels are a periodic substrate of amplitude set by V. Every configuration in this article is a displacement field $u(x)$ of electrons against kernels: the rigid slide is its uniform mode, and a kink is the localised profile carrying u from 0 to one spacing. In that model the width of the kink and the barrier to moving it are

$$W = \sqrt{\kappa/K}, \qquad K = V''(u_{\min}), \qquad E_{\mathrm{PN}} \sim \exp(-c\,W/a), \tag{3}$$

so *the barrier is exponentially sensitive to a ratio of two curvatures.* That is the worst possible form for direct measurement — an exponential magnifies every error in κ and K — and it is why differencing total energies was never going to work, whatever the mesh. It also indicates the route: κ and K are local curvatures, taken between configurations differing infinitesimally, where discretisation errors cancel instead of competing. Measuring those and inferring E_{PN} from (3) is a different calculation from the one attempted here, and it is left to a separate study. A caution belongs with it: (3) assumes nearest-neighbour coupling, while Coulomb is long-ranged, which changes kink profiles from exponential to power-law and shifts the transition. It is a guide to the regime, not a formula to trust to three figures.

What this says about scope. Less than this article set out to establish, and in a different direction. No activation energy is available, so the construction is neither placed in the hopping-semiconductor class nor excluded from it by measurement. The scope claim of §2 stands or falls on the structural argument instead — non-overlap, a vanishing transfer integral, and the absence of any electron equation of motion — which is where it should have rested from the beginning, since those follow from the postulate and need no grid at all.

Earlier figures, and why they are not quoted. A series of barrier scans was made before the difficulty above was understood, all of them by the free-boundary route: a compression test on a ring, a doping series at two concentrations, a lattice barrier between interstitial pockets of order 5 eV, and a localised interface defect costing 179 meV independently of chain length. Every one specified or evolved a partition that no symmetry fixes, so each is a bound of its stated order rather than a determination, and all are superseded by the barrier above. What survives from them is the direction of the density trend, which the symmetry-fixed measurement confirms.

What is actually transported. The carrier in this framework is not an electron. An electron here is not a particle but a unit of charge density on a domain, and what exists is the *partition* of space; dynamics is the evolution of that partition, and transport is the partition pattern travelling. What moves along the chain is the **free boundary** — a locus where two densities balance — and nothing material traverses it at all.

The quantitative form of this is the useful one. As a kink passes, each electron advances by exactly one site spacing and then stops, while the kink advances the entire length of the chain. The carrier's displacement and its constituents' displacement are decoupled, and their ratio is the chain length. This is the relation between a dislocation and the slip it produces: the dislocation crosses the crystal, no atom moves far. Charge is genuinely transported and real charge arrives at the far end; it is the *carrier* that is immaterial, in the sense that a wave is immaterial while the water is not.

That diagnosis also accounts for the failures recorded earlier in this study rather than excusing them. Each of the unsuccessful constructions imposed a carrier that had to *translate*, moving bodily through occupied space, and returned costs of tens of electron-volts or else collapsed. Those were the price of forbidding the only mechanism the framework has. The solver states as much directly: its free boundary evolves by the density balance $(\rho_i - \rho_j)/(\rho_i + \rho_j)$, flipping ownership cell by cell. Boundary motion is the only motion it performs. The corona result is the same statement seen from the favourable side — an added electron spread into a uniform shell at no cost, which is a boundary redistributing freely.

This raises the stake of the kink measurement. If transport works here it works *because* of the free boundary, which is the single feature separating this framework from the standard theory; and if the boundary cannot carry charge cheaply, the difficulty is not in a detail of the construction but in the framework's defining object.

A repair that is not available, and one that is. An obvious response to the barrier is to let valence electrons share a territory, which removes it by construction. The $n^{5/3}$ scaling rules that out: sharing destroys the extensivity the partition supplies, and a metal would lose its kinetic energy density by fifteen orders of magnitude.

What the framework then is. Non-overlapping sites, occupation 0, 1 or 2, an on-site repulsion U, and transfer between neighbours: that is the **Hubbard model**. Which locates RealQM exactly. As formulated — one unit of charge per domain, always — it is the Hubbard model in the limit $U \to \infty$, where double occupancy is forbidden outright and every site is singly occupied. That limit is a Mott insulator at every density, which is precisely what the earlier barrier scans found by direct computation, and it explains the density trend as well: no finite U can be beaten by a bandwidth that does not exist.

The repair is correspondingly minimal. Keep the partition, keep the geometry, keep the Coulomb energetics, and allow occupation to take the value two at a computed cost. The framework then sits where the standard treatment of strongly correlated matter sits — the Hubbard model is the working description of Mott insulators, of the cuprates, and of the materials for which band theory fails — with the difference that its U and its lattice are derived rather than supplied.

We do not carry this out here. What is established is where the obstruction lies: not in the interface condition, not in the geometry, but in the rule of one unit of charge per domain. That rule is also the source of one of the framework's results, charge quantization as a consequence of indivisibility rather than an assumption. The tension is worth stating plainly: *the postulate that explains why charge comes in units is the postulate that forbids a metal.* Matter has it both ways, quantizing the particles while leaving the amplitude on each site continuous, and a partition of real densities has no evident means to do the same.

6 The same chain, computed the ordinary way

Everything above is internal to the framework. It is worth putting the same system beside the standard treatment, on quantities both can produce, and the result is a check on the scope claim of §2 rather than a contest.

Kohn–Sham DFT, PBE/6-31G, all electrons, a straight chain of lithium atoms at the spacings used above, with the geometry held uniform so that the comparison is like for like — a Peierls distortion is not permitted on either side:

	RealQM	PBE/6-31G	experiment
equilibrium spacing (a_0)	4.5	6.0	5.7 (bcc n.n.)
verdict at $a = 6.0$	pinned, 184 meV	metallic	lithium conducts
verdict at its own equilibrium	depinned, ≈ 0	metallic	—
cohesive energy (eV/atom)	~ 1.7	0.486	1.63 (bulk, 8 n.n.)

And it is not one functional's answer. Density-functional theory is not the Schrödinger equation: it replaces the interacting problem with fictitious non-interacting electrons in an effective potential, exact in principle by Hohenberg–Kohn but resting in practice on a *constructed* exchange–correlation functional and a finite basis — two uncontrolled approximations. The comparison above is therefore model against model, not model against exact, and would normally be qualified accordingly. On the quantity at issue it need not be. Repeating the scan with Hartree–Fock, which carries no correlation whatever, and with MP2, which carries it perturbatively, moves the total energy by some 0.3 Ha between methods and leaves the equilibrium spacing at 6.0 a_0 in all three:

a (a_0)	HF	PBE	MP2
4.5	−51.97126	−52.29669	−52.02861
5.0	−52.01844	−52.33386	−52.06750
6.0	**−52.04726**	**−52.34332**	**−52.08273**
7.0	−52.04089	−52.31653	−52.06476

The twenty per cent is thus a disagreement with standard quantum chemistry rather than with a choice of functional.

But the objection cuts the other way, and harder. If one may choose the approximation to suit the problem then one will always win, and that is not a comparison. Applied evenly, the charge falls on *this* side of the table. The standard column used a non-empirical functional, a general-purpose basis, and three treatments of correlation that were not selected after the fact — but the RealQM column carries a free parameter, the pseudopotential radius r_c, and it was set to 1.6 a_0 by choice rather than by any independent criterion. At $r_c = 2.4$ the equilibrium spacing of the same chain moves out beyond 6.0 a_0, which *agrees* with the standard result. The twenty per cent is therefore a statement about $r_c = 1.6$ and not about the framework, and the framework could have been made to agree as easily as to disagree.

A comparison is only worth making if the parameters are fixed by something other than the quantity being compared. The proper protocol is to calibrate r_c on the *isolated* lithium atom — requiring the single pseudo-atom to reproduce the measured first ionisation energy, 0.198 Ha — and then to compute the chain with that value and no further adjustment. The chain is then a prediction rather than a fit, and stands on the same footing as the standard column, in which nothing was tuned to lithium chains. Until that is done the table above should be read as establishing the *method-independence of the standard answer* and nothing about the size of any discrepancy.

The metallic verdict is not read off a single gap, which any finite chain has, but off its scaling: 0.432, 0.346, 0.291, 0.247, 0.173 eV at $n = 5, 7, 9, 11, 13$, with $n \times$ gap constant near 2.4 eV, so the gap closes as $1/n$ and is finite-size rather than a band gap.

That verdict is weaker than it looks, and in a direction that matters. What was compared are Kohn–Sham eigenvalue differences, and Kohn–Sham eigenvalues are not excitation energies. They belong to a fictitious system of *non-interacting* particles introduced to reproduce a density; the true gap differs from theirs by the derivative discontinuity of the exchange–correlation functional, which PBE does not possess at all. Approximate functionals are known to underestimate gaps for exactly this reason, often by a factor of two. A small Kohn–Sham gap is therefore what the method returns whether or not a gap is present.

The bias has a direction, and it is the one at issue here. Approximate functionals carry a self-interaction error — an electron repelled in part by its own density — whose consequence is **delocalisation error**: densities spread too far, localised solutions are penalised, and metallic answers are favoured. So the standard column calling this chain metallic is that treatment erring in its characteristic direction, not a neutral measurement. Against it, the framework of this article over-localises, by construction, being $U \to \infty$. *Both are biased and in opposite senses*, and lithium is simply a case where the standard bias does no damage, because the chain really is metallic. On a system where it does damage — an oxide, a doped Mott insulator, anything whose gap is on-site repulsion — the delocalisation error is the well-known failure and the localisation is the correct instinct.

Two further asymmetries belong in the record. Density-functional theory admits no convergence path: wavefunction methods form a hierarchy, Hartree–Fock through MP2, CCSD, CCSD(T) to full configuration interaction, which converges to the exact answer within a basis, whereas no procedure refines PBE towards the true functional. And the quantities are not of the same kind. What is computed here are total energies of *specified configurations* — differences between states of one system, which are observables — where the gap above is an eigenvalue difference in an auxiliary system that was never claimed to have physical states. Where this framework is less accurate it is at least computing something that means what it says.

What this establishes, and what it does not. On this system the standard treatment is better and cheaply so: the equilibrium spacing within a few per cent of the measured nearest-neighbour distance, the right conduction verdict, a sensible cohesive energy, in seconds. RealQM places the atoms some twenty per cent too close, and at the spacing lithium actually adopts it predicts a pinning barrier of 184 meV where the metal has none.

That is not a surprise and should not be read as one, because *lithium is outside the scope this article claims.* §2 excludes metals in kind and not by degree: extended Bloch states, band transport, a Fermi surface, none of which the construction possesses. A simple metal is the standard treatment's best case and this framework's declared worst, so the table is a scope check and not a competition.

As a scope check it passes, in the specific sense that matters: the framework errs by *over-localising*, in bond length and in pinning alike, which is what $U \to \infty$ implies and therefore the direction the identification predicts. It fails as it was said it would, rather than arbitrarily.

The carrier itself, against exact diagonalisation. The sharpest form of the comparison does not use energies at all. Take the same object this article hops — n protons, $n-1$ electrons, nuclei clamped — and diagonalise it exactly in a minimal basis. The Loewdin site populations then say directly whether the carrier is localised:

a (a_0)	E_{corr} (eV)	$4t$ (eV)	max hole/site	
3.0	-6.43	11.11	0.223	delocalised
4.0	-11.97	4.65	0.215	delocalised
4.5	-14.64	2.96	0.226	delocalised
6.0	-19.86	0.71	0.253	delocalised

A hole spread evenly over seven sites reads 0.143; one sitting on a single site reads 1.000. At every spacing the exact ground state puts it on four or five sites. *There is no localised carrier, and therefore no barrier for one to cross.* Note that this is not because correlation is weak — E_{corr} reaches 19.9 eV — but because strong correlation and localisation are different things: a one-dimensional Mott insulator doped with a single hole conducts, the holon propagating freely, which is what the calculation reproduces.

So the disagreement with this construction is about the state and not about a number, which is the harder kind to explain away. Two qualifications keep it honest. A minimal basis is not the exact solution of Schroedinger's equation but exact diagonalisation within a deliberately crude span — STO-6G misses the hydrogen atom by six per cent before any chemistry begins — and attempts to repeat the calculation in larger bases did not converge for the stretched chains. So no exact reference exists for these configurations, and nothing here can declare RealQM's pinning wrong for them. What can be said is that two independent approximate treatments and the known behaviour of the one-dimensional Hubbard model all point the same way, and this construction points the other.

Where the comparison would actually bite. The corollary is uncomfortable and is recorded plainly: nothing computed here is a quantity the standard treatment does not already supply more accurately. The case for the construction rests instead on the real-space mechanism — a registry, a sliding partition, a carrier that is a boundary rather than a particle — being a more transparent account of activated hopping than a model Hamiltonian with fitted parameters. That claim is not tested by lithium. It would be tested on the systems the scope table names, where on-site repulsion rather than band structure sets the gap and where density-functional theory is itself in difficulty: a doped Mott insulator, an organic semiconductor, a polaronic oxide. Those comparisons are not made here.

A note on what was tested numerically

The plasma frequency $\omega_p^2 = 4\pi n$, the screening relation, the Thomas–Fermi comparison and the vanishing of C are analytic. The alkali lattice constants are a radial Wigner–Seitz calculation with the stated pseudopotential and correlation functional, run for five alkalis; the standard column reproducing the series is what makes the comparison meaningful, and its own 44% mean error in the bulk modulus sets the accuracy available at this level of model. Three attempts are reported as null rather than omitted. The interface parameter β was tested on helium at three meshes at fixed imaginary time, giving shifts of $+0.151$, -0.018 and -0.184 Ha as the mesh refined — swinging through zero and growing, with absolute energies moving away from the exact -2.9037. The cause is the geometry rather than the arithmetic: in helium the two domains meet at a plane *through the nucleus*, so interface error and Coulomb-cusp error are superimposed and cannot be separated on a uniform grid. The measurement wants H_2, where the domains meet at the midplane *between* the nuclei in a smooth region. An interface-condition test on helium was not converged, giving a sensitivity that reversed sign under mesh refinement, and no conclusion is drawn from it. Two chain calculations — five hydrogens with an excess electron, and five hydrogen kernels with one electron missing — returned no transport, but for a reason established by reading the constructions rather than by the runs: a carrier without a kernel feels no lattice force, and a bare kernel receives no domain for a vacancy to occupy. Both are recorded because the null results located those obstructions. A caution on the numerics belongs with them. Several attempts returned energies below the physical floor for their particle content, but these track the boundary speed and the seeding rather than the configuration: with the interface driven faster than the densities can follow, or with a carrier seeded directly onto an occupied kernel, the relaxation runs away, while the same configurations at a quarter of that speed settled to physical values. A two-domain shell on one nucleus is in fact stable under the density-balance rule — compressing the inner domain raises its density at the interface and lowers the outer's, which pushes the boundary back out — and both domains carry real kinetic energy there, since the nucleus shapes their profiles. The vanishing of C is a statement about a *uniform* density and does not transfer to a shaped one. Scripts and pages accompany the source.

7 Conclusion

The question was whether a construction of non-overlapping unit charge densities, meeting at free boundaries, conducts. The answer is that in this formulation it cannot be computed, and the reasons are worth more than the number would have been.

The carrier is a free boundary, and that identification holds. An electron here is a unit of charge density on a domain; what exists is the partition of space. Transport is therefore the partition pattern travelling, in the relation a dislocation bears to slip: charge arrives at the far end while nothing material crosses the sample. That picture accounts for every null result reported above — each had imposed a carrier obliged to translate bodily through occupied space — and it is why the one transport calculation this framework performs successfully is Grotthuss conduction, where the carrier is a proton and the framework does have an equation of motion for it.

The picture has a definite limit, and it should be stated with the claim rather than after it. Defect-mediated transport is real — dislocation glide, charge-density-wave sliding and ionic conduction all proceed that way — but it is not how a metal conducts. Residual resistivity *falls* as defects are removed, so a carrier that must migrate as a defect predicts the opposite of what purity does. The free boundary is therefore the mechanism this construction is restricted to, not the carrier of ordinary conduction, and its prominence here is a fact about the formalism rather than a discovery about charge transport.

Costing that motion defeats the method, and the control is what shows it. A barrier of 0.01 Ha differenced from totals of 1.3 Ha carrying 0.27 Ha of first-order discretisation error requires a cancellation it does not get, and brute force cannot supply it: reaching ten per cent of the barrier needs a 42000^3 grid. Scanning a full lattice period instead of sampling two configurations supplies the diagnosis for free, since translation by one spacing must return the same energy. Six of eight such scans fail that control, the two that pass survive no change of parameter, and the only scan at parameters the framework does not choose fails by the widest margin. What the scans do contain is a drift of the carrier toward a chain end and grid-locking of a staircase interface — both of order an electron-volt, both exactly reproducible, which is why they were mistaken for signal for as long as they were.

Beneath that lies a structural obstacle no refinement touches. Non-overlap makes the transfer integral vanish identically, so band conduction is absent in kind: there are no extended states to carry it. And in this scheme nuclei move under classical equations of motion while electrons relax to a static minimum — so there is no electron dynamics, and hence no object in the theory corresponding to a current. *A minimisation has no time in it.* Every attempt above infers transport from an energy landscape, which is a proxy, and it is the proxy that breaks. The missing ingredient is not a better barrier calculation but an equation of motion, and it exists elsewhere in the programme: the time-dependent form in which a real density carries a complex current is where a conduction calculation belongs.

What the framework is, and is not. Single occupancy with long-range Coulomb is *not* the Hubbard atomic limit, whose degeneracy over carrier position would give a barrier of exactly zero; the Coulomb interaction between extended densities breaks that degeneracy, placing the construction closer to an extended Hubbard model where charge ordering is genuine physics. Within bound matter — structure, binding, geometry — it works, and the transport regimes it reaches are those with nuclear carriers. Electronic transport is outside, and against exact diagonalisation on the same chains the reason is visible directly: the exact carrier occupies four or five sites where this construction admits only one.

The result is therefore a boundary, not a barrier. A method with a mapped edge is more usable than one claimed to be universal, and the edge established here is sharp: static RealQM describes bound matter and nuclear transport, and cannot represent electronic conduction at all. Everything this article originally set out to measure lay on the far side of it.

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